

1. Big-O Notation

$$f_1 = n^2 + n \log n + 3, \quad f_2 = 32(n+1)(n-1), \quad f_3 = n^2 \log^2 n, \quad f_4 = \frac{5n^2}{\log n} (\log n + 2)^3, \quad f_5 = \sqrt{n}(n^{3/2} + 2)$$

$$f_1 = n^2 + n \log n + 3 \in \Theta(n^2)$$

$$f_2 = 32(n^2 - 1) \in \Theta(n^2)$$

$$f_3 = n^2 \log^2 n \in \Theta(n^2 \log^2 n)$$

$$f_4 = 5n^2 \log^2 n + 30n^2 \log n + 60n^2 + \frac{40n^2}{\log n} \in \Theta(n^2 \log^2 n)$$

$$f_5 = n^2 + 2\sqrt{n} \in \Theta(n^2)$$

$$(f_1, f_2), (f_1, f_5), (f_2, f_5) \in \Theta(n^2) \quad (f_3, f_4) \in \Theta(n^2 \log^2 n)$$

2. Ranking Asymptotic Growth Rates

$$n^4, \quad 2^{\sqrt{\log n}}, \quad n \log^2 n, \quad 2^n, \quad (n+1)!, \quad 2^n n!, \quad n 2^n, \quad n \log n$$

$$\log(2^{\sqrt{\log n}}) = \sqrt{\log n} = o(\log n) \implies 2^{\sqrt{\log n}} = o(n^\epsilon)$$

$$\frac{n \log n}{n \log^2 n} = \frac{1}{\log n} \rightarrow 0$$

$$\frac{n^4}{2^n} \rightarrow 0 \quad (\text{polynomial} < \text{exponential})$$

$$\frac{(n+1)!}{2^n \cdot n!} = \frac{n+1}{2^n} \rightarrow 0$$

$$2^{\sqrt{\log n}} < n \log n < n \log^2 n < n^4 < 2^n < n 2^n < (n+1)! < 2^n n!$$

3. Analysis of Snippets

(a)

begin

$M \leftarrow 0$

$i \leftarrow 1$

while $i \leq n$ **and** $M < A[i][i]$ **do**

for $j \leftarrow i$ **to** n **do**

if $M < A[i][j]$ **then**

$M \leftarrow A[i][j]$

$i \leftarrow i + j$

After for-loop: $j = n + 1$

$i \leftarrow 1 + (n + 1) = n + 2 > n$

While-loop iter.: ≤ 1

For-loop iter.: $= n$

$$T(n) = O(n)$$

(b)

```

begin
   $M \leftarrow 0$ 
   $i \leftarrow 1$ 
  while  $i \leq n$  do
     $j \leftarrow 2$ 
    while  $j \leq n$  do
      if  $M < A[i][j]$  then then
         $M \leftarrow A[i][j]$ 
       $j \leftarrow j + i$ 
     $i \leftarrow i + 1$ 

```

Outer: $i = 1, \dots, n \quad (n)$ Inner: $j = 2, 2+i, \dots \quad (n/i)$

$$T(n) = \sum_{i=1}^n \frac{n}{i} = nH_n$$

$$T(n) = \Theta(n \log n)$$

(c)

```

begin
   $M \leftarrow 0$ 
   $i \leftarrow 1$ 
  while  $i \leq n$  do
     $j \leftarrow 2$ 
    while  $j \leq n$  do
      if  $M < A[i][j]$  then then
         $M \leftarrow A[i][j]$ 
       $j \leftarrow j^2$ 
     $i \leftarrow 2 \cdot i$ 

```

Outer: $i = 1, 2, 4, \dots, 2^k$ $(k = \lfloor \log_2 n \rfloor)$ Inner: $j = 2, 4, 16, \dots$ $(m = \lfloor \log_2 \log_2 n \rfloor)$

$$T(n) = O(\log n \log \log n)$$

(d)

```

begin
   $M \leftarrow 0$ 
   $i \leftarrow 1$ 
  while  $i \leq n$  do
     $j \leftarrow 2$ 
    while  $j \leq n$  do
      if  $M < A[i][j]$  then then
         $M \leftarrow A[i][j]$ 
       $j \leftarrow i + j$ 
     $i \leftarrow 2 \cdot i$ 

```

Outer: $i = 2^0, 2^1, \dots \quad (O(\log n))$ Inner: $O(n/i)$

$$T(n) = \sum_{k=0}^{\log n} \frac{n}{2^k} < 2n$$

$$T(n) = O(n)$$

4. Interval Scheduling

(a)

$$I_1 = (1, 3), \quad I_2 = (2, 5), \quad I_3 = (4, 7), \quad I_4 = (6, 9), \quad I_5 = (7, 8)$$

Conflicts: $\{I_1, I_2\}, \{I_2, I_3\}, \{I_3, I_4\}, \{I_4, I_5\}$ Graph: $I_1 - I_2 - I_3 - I_4 - I_5 \quad (P_5)$

$$S^* = \{I_1, I_3, I_5\}, \quad B_1 = 3 \leq 4 = A_3, \quad B_3 = 7 \leq 7 = A_5$$

$$\forall S \subseteq \{I_1, \dots, I_5\}, \quad |S| = 4 \implies \exists \text{ conflict}$$

$$\boxed{S^* = \{(1, 3), (4, 7), (7, 8)\}, \quad |S^*| = 3}$$

(b)

$$I_1 = (1, 5), \quad I_2 = (0, 2), \quad I_3 = (4, 6)$$

$$I_2 \cap I_3 = \emptyset, \quad I_1 \cap I_2 \neq \emptyset, \quad I_1 \cap I_3 \neq \emptyset$$

$$\boxed{S_{\text{maximal}} = \{I_1\}, \quad |S| = 1 \quad S_{\text{maximum}} = \{I_2, I_3\}, \quad |S| = 2}$$

(c)

$$I_k = (k, k + 6), \quad k = 0, \dots, 5 \quad \forall i \neq j : I_i \cap I_j \ni 5 \implies K_6$$

$$\boxed{\text{Max size} = 1, \quad \text{Optimal solutions} = 6}$$

(d)

$$\text{G1: } I_1 = (0, 6), \quad I_2 = (2, 8), \quad I_3 = (4, 10) \quad \text{G2: } I_4 = (10, 16), \quad I_5 = (12, 18), \quad I_6 = (14, 20)$$

$$\forall i, j \in G_k, \quad i \neq j : I_i \cap I_j \neq \emptyset \quad \forall i \in G_1, \quad j \in G_2 : B_i \leq 10 \leq A_j$$

$$\boxed{\text{Max size} = 2, \quad \text{Optimal solutions} = 3 \times 3 = 9}$$